

Enhancing a Context-Aware Backlash Risk Analyzer Using Self-Consistency

Seungjong Yoo¹

¹Department of Computer Science, Soongsil University, Seoul, South Korea

ABSTRACT

This paper presents a Context-Aware Backlash Risk Analyzer that estimates potential social backlash across different communication contexts. Existing sentiment analysis methods often fail to capture context-dependent risks such as sarcasm, aggression, and group generalization. To improve assessment reliability, we introduce a self-consistency mechanism that aggregates multiple LLM evaluations and produces a final risk score with an uncertainty estimate. Experimental results show that self-consistency reduces output variability and provides more stable risk assessments. These findings suggest that self-consistency is an effective approach for improving the robustness and interpretability of context-aware social risk analysis systems.

1. INTRODUCTION

Online communication has become a major part of everyday life through social media platforms, online communities, messaging applications, and workplace communication tools. While these platforms make communication faster and more accessible, they also increase the possibility of misunderstandings, negative reactions, and social backlash. A message that appears harmless in one situation may be interpreted as offensive, inappropriate, or controversial in another context. As a result, understanding the potential social impact of written text has become an important challenge.

Traditional sentiment analysis focuses primarily on identifying whether a text expresses positive, negative, or neutral sentiment. However, social backlash is often influenced by more complex factors than sentiment alone. Expressions involving sarcasm, aggression, stereotypes, excessive confidence, or contextually inappropriate language may trigger negative responses even when the overall sentiment is not strongly negative. This limitation makes conventional sentiment analysis insufficient for assessing real-world communication risks.

Recent advances in large language models (LLMs) have enabled more sophisticated text understanding and context-aware reasoning. These models can evaluate subtle linguistic patterns and consider different communication contexts when interpreting a message. Nevertheless, LLM-based assessments often suffer from output variability, where repeated evaluations of the same input may produce slightly different results. Such inconsistency reduces the

reliability of risk assessment systems.

To address this issue, this project extends a Context-Aware Backlash Risk Analyzer by incorporating a self-consistency mechanism. Instead of relying on a single evaluation, the system performs multiple independent assessments and aggregates the results to obtain a more stable risk estimate. The objective of this study is to investigate whether self-consistency can improve the robustness and interpretability of context-aware social backlash risk analysis while maintaining practical usability.

2. RELATED WORK

2.1. Context in Toxicity Detection

Pavlopoulos et al. investigated whether conversational context influences toxicity judgments in online discussions [1]. Their study demonstrated that toxicity perception can change when additional conversational context is provided. The authors showed that contextual information may either increase or decrease the perceived toxicity of a comment, highlighting the importance of context in content moderation tasks. These findings suggest that evaluating text in isolation can lead to inaccurate judgments. However, their work primarily focused on toxicity classification and did not address broader forms of social backlash risk such as sarcasm, stereotyping, overconfidence, or contextual inappropriateness.

2.2. Contextual Abuse Detection

Vidgen et al. introduced the Contextual Abuse Dataset (CAD), a large-scale dataset designed for abuse

detection with conversational context [2]. Unlike previous datasets that labeled individual comments in isolation, CAD incorporates discussion threads and contextual information during annotation. The dataset demonstrated that abusive language is often highly dependent on surrounding context and cannot always be identified from a single message alone. This work motivates the use of context-aware approaches in social risk analysis. Nevertheless, the primary objective of CAD is abuse classification rather than estimating the overall likelihood of social backlash across different communication environments.

2.3. Self-Consistency for Reliable LLM Reasoning

Wang et al. proposed Self-Consistency, a decoding strategy that improves reasoning performance in large language models by sampling multiple reasoning paths and selecting the most consistent final answer [3]. Their experiments showed significant performance improvements across arithmetic and commonsense reasoning benchmarks. The key insight is that aggregating multiple independent reasoning processes can reduce the effect of stochastic model outputs and increase reliability. Inspired by this idea, our work applies self-consistency to social backlash risk assessment. Instead of selecting a single evaluation result, multiple independent assessments are aggregated to generate a more stable risk score and an uncertainty estimate.

2.4. Summary

Previous studies have demonstrated the importance of contextual understanding in toxicity and abuse detection, while recent advances in self-consistency have improved the reliability of LLM-based reasoning. Building upon these findings, this project combines context-aware social risk analysis with a self-consistency mechanism to improve the robustness and interpretability of backlash risk assessment.

3. METHODS

3.1. System Overview

The proposed system is a Context-Aware Backlash Risk Analyzer designed to estimate the likelihood of social backlash from a given text. Unlike traditional sentiment analysis systems that focus primarily on positive or negative sentiment, the proposed approach evaluates multiple dimensions of social risk while considering the communication context in which the message is delivered.

The complete source code is publicly available on GitHub [4].

The overall workflow consists of four stages: data preprocessing, context-aware risk evaluation, self-consistency aggregation, and final risk score generation.

3.2. Data Preprocessing

Before evaluation, input texts are normalized to reduce unnecessary variation. The preprocessing stage includes whitespace normalization, text cleaning, and basic formatting standardization. The processed text is then paired with contextual information describing the communication environment.

Examples of communication contexts include:

- Public Social Media
- Workplace Email
- Private Messaging
- Online Discussion Forum

The context description is incorporated into the evaluation prompt so that the same statement can be interpreted differently depending on the communication scenario.

3.3. Risk Evaluation Model

The system employs a Large Language Model (LLM) as the primary evaluator. Instead of performing simple sentiment classification, the model estimates potential backlash risk through multiple dimensions associated with negative social reactions.

The evaluated dimensions are:

- Aggression
- Group Generalization
- Sarcasm or Mockery
- Overconfident Judgment
- Context Inappropriateness
- Misinterpretability
- Norm Violation

Each dimension contributes to the final backlash risk score generated by the system.

3.4. Self-Consistency Mechanism

The original baseline system generated a single evaluation for each input text. However, due to the probabilistic nature of LLMs, repeated evaluations of the same text may produce slightly different outputs. To improve reliability, a self-consistency mechanism was introduced. For each input sample, the system performs five independent evaluations using identical prompts. The resulting scores are aggregated to obtain a more stable estimate.

Let s_i denote the risk score from the i -th evaluation and N denote the number of evaluations. The final risk score is computed as:

$$FinalScore = \frac{1}{N} \sum_{i=1}^N s_i \quad (1)$$

To quantify output stability, the standard deviation of the scores is calculated as an uncertainty estimate:

$$Uncertainty = \sqrt{\frac{1}{N} \sum_{i=1}^N (s_i - \bar{s})^2} \quad (2)$$

where $\bar{s}$ represents the mean risk score.

A lower uncertainty value indicates that the model produced more consistent judgments across multiple reasoning paths.

3.5. Experimental Setup

Experiments were conducted using a context-sensitivity benchmark consisting of text samples evaluated under multiple communication contexts. The same benchmark was analyzed using both the baseline system and the upgraded self-consistency system.

Table 1 summarizes the experimental configuration.

Table 1. Experimental Configuration

Item	Value
Evaluation Method	LLM-based
Communication Contexts	Multiple
Risk Dimensions	7
Self-Consistency Runs	5
Output Metrics	Risk Score, Uncertainty

3.6. Result Analysis

The effectiveness of the proposed approach was evaluated by comparing the baseline and self-consistency systems.

The following metrics were analyzed:

- Mean Risk Score
- Risk Score Difference
- Output Variability
- Mean Uncertainty

The experimental results were visualized using tables and graphs. Particular attention was given to whether self-consistency reduced score fluctuations and improved the stability and interpretability of context-aware backlash risk assessment.

4. RESULTS

To evaluate the effectiveness of the proposed self-consistency mechanism, the baseline system was

compared with the upgraded self-consistency system. The comparison focuses on the average risk score, score difference, and uncertainty estimate.

Table 2 presents the overall comparison. The baseline system produced an average risk score of 40.87, whereas the self-consistency system produced an average risk score of 38.67. This indicates a reduction of approximately 2.20 points. The upgraded system also produced a mean uncertainty value of 0.09, which provides an additional indicator of output stability.

Table 2. Overall Comparison Between Baseline and Self-Consistency

Metric	Baseline	Self-Consistency
Mean Risk Score	40.87	38.67
Difference	–	-2.20
Mean Uncertainty	N/A	0.09

The results indicate that self-consistency produces slightly lower risk scores while improving output stability. By aggregating multiple independent evaluations, the upgraded system reduces the influence of individual stochastic model outputs.

Table 3 shows the average risk score across different communication contexts. In most contexts, the self-consistency system produced lower scores than the baseline system. The largest reduction was observed in Email / Private, where the average score decreased from 30.2 to 25.8.

Table 3. Average Risk Score by Context

Context	Baseline	Self-Consistency
Email / Private	30.2	25.8
Email / Public	55.2	53.4
Message / Private	31.0	28.2
Message / Public	50.0	47.6
SNS / Private	31.8	29.8
SNS / Public	47.0	47.2

Figure 1 visualizes the overall difference between the baseline and self-consistency systems. The self-consistency system maintained a similar overall scoring pattern while slightly reducing the average risk score.

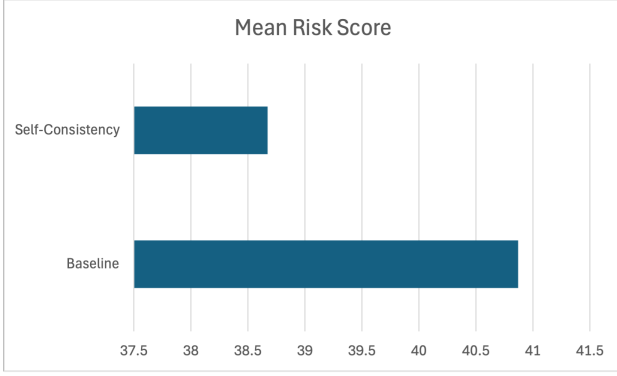


Figure 1. Average risk score comparison between the baseline and self-consistency systems.

Across most communication contexts, the self-consistency mechanism generated lower and more stable risk estimates. This suggests that repeated evaluation can reduce score fluctuations, especially in less explicit social-risk scenarios. Overall, the experimental findings show that self-consistency improves the robustness of context-aware backlash risk assessment while maintaining similar overall evaluation behavior.

5. DISCUSSION

The experimental results indicate that incorporating self-consistency improves the stability of context-aware backlash risk assessment. While the average risk scores generated by the upgraded system remain similar to those of the baseline system, the self-consistency mechanism consistently reduces score variability by aggregating multiple independent evaluations.

The reduction in average risk score suggests that individual model outputs may occasionally overestimate backlash risk due to stochastic variations in LLM reasoning. By averaging multiple evaluations, the upgraded system produces more balanced and reliable estimates. This effect was particularly noticeable in private communication contexts, where social risk signals are often less explicit and more sensitive to interpretation.

Another important contribution of the proposed approach is the introduction of an uncertainty estimate. Rather than providing only a single risk score, the system also reports how consistent the evaluations are across multiple reasoning paths. This additional information can help users better understand the confidence of the assessment and identify cases where the model may be uncertain.

Despite these improvements, several limitations remain. The evaluation benchmark is relatively small and does not fully represent the diversity of real-world

online communication. Furthermore, the system relies on a single LLM, making its performance dependent on the capabilities and biases of the underlying model. Future work may explore larger evaluation datasets, alternative aggregation strategies, and the use of multiple language models to further improve reliability and generalizability.

6. CONCLUSION

This study presented an enhanced Context-Aware Backlash Risk Analyzer that incorporates a self-consistency mechanism for improving the reliability of LLM-based social risk assessment. Unlike traditional sentiment analysis approaches, the proposed system evaluates multiple dimensions of backlash risk while considering the communication context in which a message is delivered.

Experimental results showed that self-consistency reduced output variability and generated more stable risk estimates across different communication scenarios. The introduction of an uncertainty estimate further improved the interpretability of the assessment by providing information about evaluation consistency.

Overall, the findings suggest that self-consistency is an effective and practical approach for improving the robustness of context-aware backlash risk analysis. Future research may focus on expanding evaluation datasets, refining risk dimensions, and exploring more advanced aggregation methods to further enhance system performance.

7. REFERENCES

- [1] J. Pavlopoulos, J. Sorensen, L. Dixon, N. Thain, and I. Androutsopoulos, “Toxicity detection: Does context really matter?” *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 2020.
- [2] B. Vidgen, D. Nguyen, H. Margetts, P. Rossini, and R. Tromble, “Introducing cad: The contextual abuse dataset,” in *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 2021, pp. 2289–2303.
- [3] X. Wang, J. Wei, D. Schuurmans, et al., “Self-consistency improves chain of thought reasoning in language models,” *International Conference on Learning Representations (ICLR)*, 2023.
- [4] S. Yoo, *Context-aware backlash risk analyzer*, <https://github.com/seungjongyoo/context-aware-backlash-risk-analyzer>, 2026.